

Faster Region-Convolutional Neural Network with ResNet50 for Video Stream Object Detection

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Abstract—Detecting objects through video frames is a difficult process due to several issues, such as poor quality of frames, pose changes, occlusions, video defocusing, and motion blur. The closely arranged frames make it more complicated to detect the object in video streams. A common strategy employed to identify the problem is to correctly detect objects through video sequences. Considering these problems, the Faster Region-Convolutional Neural Network (FRCNN) is implemented to detect objects in video streams. At first, the Region proposal network (RPN) is used to generate the objects feature, and then the Convolutional Neural Network (CNN) classifies the RPN objects' features. Then the proposed architecture is combined with the ResNet50, which is the backbone of the Faster R-CNN pipeline, defining the FRCNN-ResNet50 object detector that constant detects objects in video streams. Then the model is evaluated in the DAVIS dataset and gets the F-measure 0.893, S-Measure 0.859, and Mean Absolute Error (MAE) 0.016 respectively. Compared to recent works like CSAttention-ConvLSTM, Deep-GCN, CFCN-MA, the proposed model achieves a higher detection rate and lower false detection.

Keywords—Convolutional Neural Network, Occlusions, Object Detection, Region Proposal Network, Video Defocusing.

I. INTRODUCTION

Object identification is significantly improving due to the ongoing development in Deep Learning (DL) algorithms. Researchers are conducting various studies on detection by treating every frame of a video considered as an image to identify objects within the footage [1]. Additionally, video surveillance remains a crucial focus for enhancing the operations and stability of modern industries and urban environments [2]. The insufficient of specific datasets containing small objects is being partially addressed by the emergence of Unmanned Aerial Vehicles (UAVs) equipped with built-in cameras, facilitating the recording of wide areas with tiny objects at standard quality levels [3]. However, the limited presence of abnormalities in publicly available datasets poses a challenge for training effective classifiers [4]. The advancement of machine mechanism onto mobile devices, such as drones and robots, is enabling a wide spectrum of automated mobile vision applications, including the use of mobile devices for localization and direction. [5]. So, The Video Stream Object

Detection (VSOD) introduced in previous research to extract the visually characteristic objects in the video stream [6]. To detect the objects in videos, existing models are used to boost up the reasoning process by evaluation feature maps, and optical flow warping techniques are used to merge the feature maps, which suffers in VSOD due to the poor quality of frames, such as pose changes, occlusions, video defocusing, and motion blur [7].

- To address the motion blur problem, an implementation of the FRCNN-ResNet50 model for particular object detection
- To improve object detection in video streams, the RPN is used to lower the false detection rate.
- A ResNet50 architecture is presented to manage feature extraction using an RPN model, hence improving the detection rate.

The remainder of the paper is organized as follows: In Section 2, Related work is discussed. In Section 3, the proposed FRCNN-ResNet50 model is explained. In Section 4, the results obtained from the performed models are discussed. Section 5, conclusion of the research is given.

II. LITERATURE REVIEW

Chenchu X. et al. [8] introduced the Dual-Stream Spatiotemporal attention for object detection via video stream in SegTrack2, DAVIS, FBMS, and VOS dataset. They introduced a three-stage model structure, attention-augmented convolutional Long Short-Term Memory (LSTM) model. And the local pixel correlation structure of every pixel in the stream object and the nearby object were done with attention-based 1D dilated convolution. Finally, the feature by attention feature selection with the help of the refinement fusion module. However, the complexity of the approach might pose challenges in terms of practical implementation, limiting its scalability in resource-constrained environments.

Mingzhu X. et al. [9] introduced the Multi-Stream Attention-Aware Graph Convolution Network (GCN) for object detection in video streams with DAVIS, ViSal, and UVSD datasets. Edge-gated Graph Convolution (GC) was performed on the graph data to increase the primary information total. Additionally, there

was a module for attention that used adaptive node selection to describe Spatiotemporal Semantic Information (SSI). Finally, smoothness-aware regularization was used to improve parallel stream objects. However, the computational demands might limit the real-time applicability of the approach, especially when dealing with large-scale video datasets.

Qingping Z. et al. [10] presented real-time video stream object detection through Cascaded Fully Convolutional Network with Motion Attention (C-FCN)-MA on the FBMS, ViSal, DAVSOD, and DAVIS datasets. Then, to improve the prediction accuracy of the initial FCN in the flow motion attention mechanism, the FlowNet2-SD model was used for effective and efficient motion pattern recognition. However, the FlowNet2-SD model for motion pattern recognition may introduce additional computational complexity, potentially making real-time video stream object detection more resource-intensive.

Yi W.C. et al. [11] introduced Contrastive Feature and Attention Modules for object detection through video stream with the DAVIS, FBMS, ViSal, and DAVSOD datasets. First, the global information capture in the video frame was done with a non-local self-attention scheme. To merge low- and high-level features through co-attention formulation, the representation of the features was improved by contrastive learning for accurate object detection. However, the non-local self-attention scheme and contrastive learning for feature representation improvement led the model into real-time applicability of the method.

Bing L. et al. [12] introduced the spatiotemporal attention enhancement with Discriminative Network (DN) for video object detection in VSOD, DAVIS, SegTrack V2, UVSD, FBMS, MCL, and ViSal datasets. A multiscale spatiotemporal ConvLSTM model was implemented to improve high-level semantic extraction. To obtain effective spatiotemporal

Characteristic's, the ConvLSTM model was combined with an attention mechanism (SCAtt-ConvLSTM) model. The Channel-wise Attention (CA) unit and the Spatial-wise Attention (SA) units, to improve the model performance. However, the integration of multiple attention mechanisms led the model to an intricate architecture, making the model challenging to interpret and optimize.

III. PROBLEM STATEMENT

- Video streams object detection affects the detection rate due to the poor quality of frames, such as pose changes, occlusions, video defocusing, and motion blur.
- The frame of the video stream is computationally difficult for detection through the video stream of each frame.

IV. OBJECTIVE

- Improve robustness in detecting stream objects in videos by developing algorithms that effectively handle challenges posed by poor frame quality, including pose variations, occlusions, defocusing, and motion blur.
- Enhance computational effectiveness in the detection of stream objects by developing a simplified strategy capable of quickly processing frames inside video streams, ensuring that the method's computational complexity is still manageable even when dealing with large amounts of video data

V. MATERIAL AND METHODS

This section presents a concise overview of the data preparation, implemented models, and performance analysis of the model used for video stream object detection. Fig. 1. Represents the work flow.

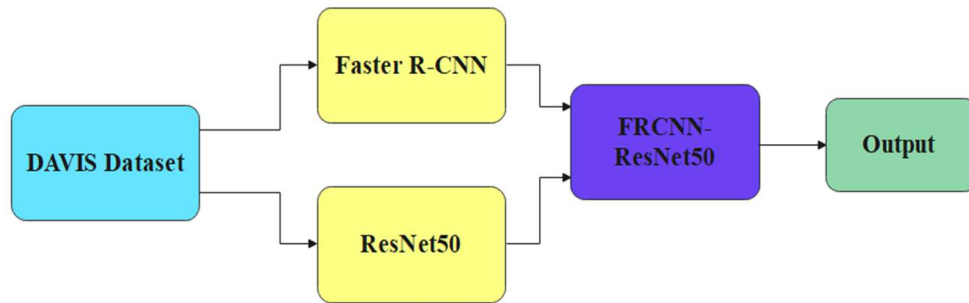


Fig. 1. Overall process of proposed word

A. Dataset

Video stream datasets, namely the Densely Annotated Video Segmentation (DAVIS) dataset, are being used to train the developed model [13]. There are 50 movies totaling 3455 frames in the DAVIS dataset. With a carefully annotated binary ground truth, each movie has around 70 frames with a resolution of 854×480 pixels. The FRCNN-ResNet50 model uses these datasets as inputs to identify objects in video streams.

B. FRCNN-ResNet50

A CNN backbone (ResNet50), a Region Of Interest (ROI) pooling layer, Fully Connected (FC) layers, and two branches for detection and bounding box regression make up the Faster R-CNN detector. The DAVIS dataset video sample is being fed into the main CNN (ResNet50) in this part in order to extract a feature map with dimensions of 8 x 40 x 512, which is the same feature-extracting feature map as the one in Region.

Proposal Network (RPN), which distributes weights along its backbones to produce object proposals, and its faster R-CNN backbone, which is used to categorize RPN proposals. The following step is pooling features from the backbone feature map using the bounding boxes that the RPN suggested. The ROI pooling layer does this by using the region from the backbone feature map that corresponds to a proposal. The region is then

split up into a predetermined number of sub-windows, and a fixed-size output is obtained by max-pooling across these sub-windows. The stream items in video frames are effectively identified by using this layer. Ultimately, characteristics are supplied into sub node detection branches after passing via two FC layers [14]. Fig. 2 represents structure of Faster R-CNN

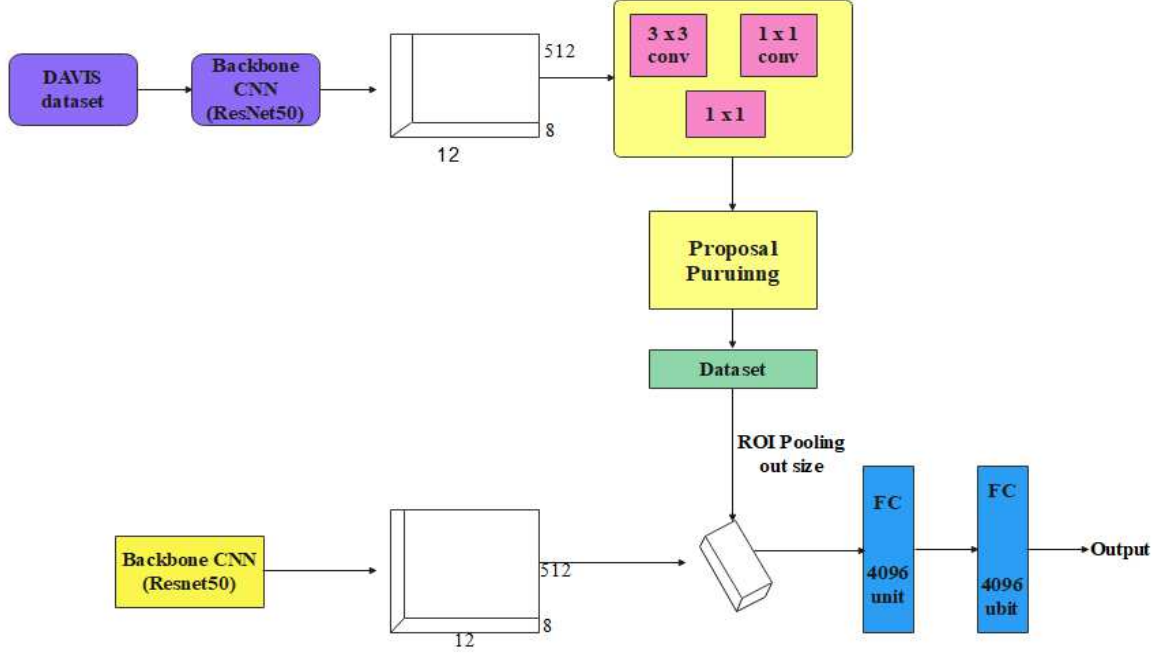


Fig. 2. Structure of Faster R-CNN

1) *Convolution Layer*: CNN represent a powerful machine training approach that processes various forms of input, including text, grids, and images. Thus, the model is implemented to learn complex patterns inherent in the entire video frame. This learning process is conducted through a channel-based convolution procedure given as (1)

$$[\lambda_{i,j}, Z_{i,j}] \times \xi_{i,j} = \lambda - \xi + 1 \quad (1)$$

The window shifts after each operation, and the features are refined through the use of feature maps. These feature maps capture the receptive field's local information in the images and facilitate shared weights and biases given as (2)

$$O_{CONV} = \sigma_{sigmoid}(b + \sum_{i=0}^2 \sum_{j=0}^2 W_{i,j} act_{a+i,b+j}) \quad (2)$$

Padding is used to preserve the dimensions of the input image. In 'SAME' padding, the resultant image size remains the same as the input image size, while 'VALID' padding is considered as no padding as (3)

$$[\lambda_{i,j}, Z_{i,j}] \times \xi_{i,j} = \frac{(\lambda + 2p - \xi)}{(\phi_s + 1)} \quad (3)$$

In this context, O_{CONV} denotes the output, p represents the padding, ϕ_s stands for the stride, b denotes the bias, $\sigma_{sigmoid}$ is

the activation function, $W_{i,j}$ refers to the weight matrix of shared weights, and $act_{a+i,b+j}$ represents the activation at position i, j .

Upon padding the resultant array, the convolutional layer acquires a corresponding feature map for the image in question. After the padding of the result, the convolution layer get an element map for the concerning image is given as (4)

$$\bar{\Omega}_{FM} = \begin{bmatrix} \lambda_{i,j} \\ Z_{i,j} \end{bmatrix}_N \quad (4)$$

2) *Region Proposal Network*: The Region Proposal Network (RPN) processes the feature map to detect the presence of objects and their movements. It employs a sliding window to identify promising regions (cells) where objects might exist. Each identified region is represented as an anchor in the form of a square in the image.

Following the RPN network prediction, a parallel number of bounding boxes is generated, with their positions and sizes adjusted. Non-max Suppression (NMS) is utilized to select box with a high foreground, passing to the subsequent level is given as (5) and (6).

$$LOSS = L(\lambda_1, \lambda_1^*, B_1, B_1^*) + L(Z_2, Z_2^*, B_2, B_2^*) \quad (5)$$

$$LOSS = [(\lambda, \lambda^*, Z, Z^*, B, B^*)] = L_{CLS}(\lambda, \lambda^*)L_{CLS}(Z, Z^*) + \lambda L_{REG}(B, B^*) \quad (6)$$

Here, λ , Z , and B denote the predicted labels for text, images, and boxes in the RPN, while λ^* , Z^* , B^* represents the true labels for text, images, and boxes. L_{CLS} and L_{REG} refer to the losses of the classifier and regressor, respectively, and λ signifies the training rate given as (7).

$$\forall_{RPN} \begin{bmatrix} B_1(\lambda_{i,j}) \\ B_2(Z_{i,j}) \end{bmatrix}_N \quad (7)$$

3) *ROI Layer*: At this point, the RPN produces images of varying shapes, necessitating the acquisition of a pooling layer to standardize the sizes. This aids in aligning the extracted features with the initial region proposal network and improves pixel segmentation results. The ROI layer further enhances model accuracy given as (8).

$$IoU = \frac{\lambda \cap \lambda^*}{\lambda \cup \lambda^*} \quad (8)$$

Following this, any region with an IoU greater than or equal to 0.5 is deemed a region of interest, while those falling below this threshold are disregarded. This procedure is iterated for all regions, and those with an IoU surpassing 0.3 are taken.

4) *Segmentation mask*: After that, a mask layer is applied, giving a segmentation mask for every region that contains an item, depending on the layer that came before it. The predicted coverage for each object in a picture is calculated by this mask

layer. Ultimately, the division cover object is levelled and added to a fully corresponding layer given as (9).

$$\forall_{flattened} = [\lambda_1, \lambda_2, \lambda_3, \lambda_4, Z_1, Z_2, Z_3] \quad (9)$$

5) *Fully Connected layer*: The input of the FC layer comes from the flattened mask layer that came before it, which is flattened and then processed by the fully connected layer. An artificial neural network (ANN)-like fully connected layer is prepared by the vector that has been flattened given as (10).

$$\forall_i^T = act(\sum_{i=1}^n \nabla_i \forall_{flattened} + N_b) \quad (10)$$

Here, N_b signifies the randomly initialized bias, ∇_i denotes the weight of the corresponding input node, and act indicates the activation function. The fully connected layer employs the SoftMax activation function to compute the probabilities of object labels within the input image.

6) *ResNet50*: To acquire semantic information from small objects, the backbone network employs ResNet-50 with residual learning units, effectively addressing the vanishing gradient problem that often arises in Deep Neural Networks (DNN). The implemented network features a deeper architecture designed to extract feature graphs rich in semantic details, all while utilizing fewer parameters and yielding more prominent effects. It helps in reducing detection faults and efficient positioning for detecting the long video stream object by deep feature extraction [15]. Fig. 3. illustrate the implemented Resnet50 process.

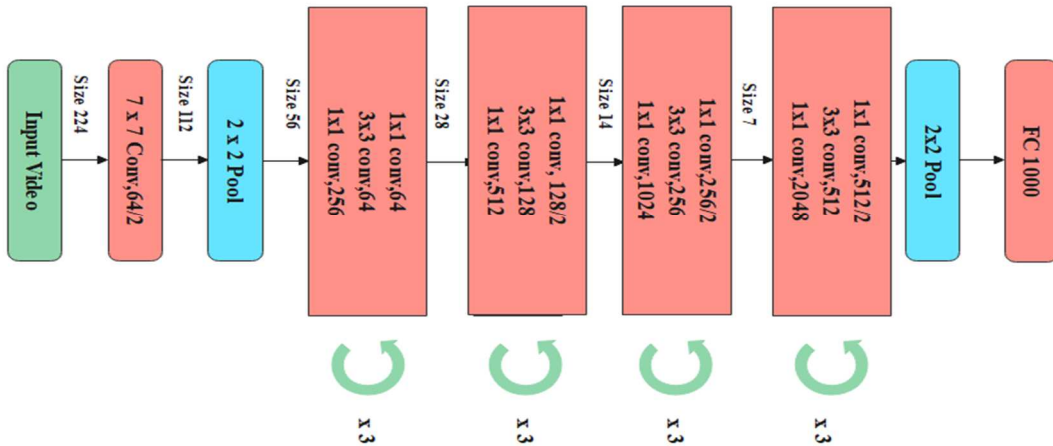


Fig. 3. Structure of ResNet50

To enhance the backbone network, the structure of ResNet50 incorporates the Conv 2_x, Conv 3_x, Conv 4_x, and Conv 5_x layers for precise feature extraction. Initially, Conv 2_x serves as the initial layer for feature extraction, while all these layers are interconnected in parallel. Lastly, the detection layer, with a dedicated unit for each of the 25 classes, utilizes a SoftMax layer to derive the detection scores. This approach mitigates the vanishing gradient problem, consequently elevating the accuracy of the detection scores in video stream object detection.

VI. RESULT AND DISCUSSION

The proposed model implementation the FRCNN-ResNet50 approach are conducted using MATLAB R2023a software. The system utilizes an Intel Core i7-13700K processor and 8GB RAM. Through appropriate power allocation, the primary objective combination of FRCNN-ResNet50 is to achieve a higher object detection rate in video streams.

A. Performance analysis

The researchers employ two assessment measures such as F-Measure and MAE, described by (11) and (12), to evaluate the performance of FRCNN-ResNet50 for video stream object detection using DAVIS and FBMS datasets.

In (11) defines the F-Measure, which is a weighted harmonic of accuracy and recall used to balance the values of true and false detection:

$$F_{\beta} - Measure = \frac{(\beta^2 + 1) \times P \times R}{\beta^2 \times P + R} \quad (11)$$

Where $\beta^2 = 0.3$ is assigned to allocate more weight to detect than recall. A set of F-Measure values is first computed for each saliency map with the threshold ranging from 0 to 255, leading to an average F-Measure score.

The S-measure, which is determined by, measures the similarity of non-binary foreground maps. It consists of an object-aware structural similarity and a region-aware structural similarity given as (12).

$$S = \alpha \cdot S_o + (1 - \alpha) \cdot S_r \quad (12)$$

Where $\alpha \in [0, 2]$ and α is here in empirically set to 0.4. The details of the computation S_o and S_r are overflowed.

The MAE metric stream and non-stream pixels to calculate the false detection rate in video stream object detection is calculated by (13):

$$MAE = \frac{1}{w \times h} \sum_{x=1}^w \sum_{y=1}^h |S(x, y) - G(x, y)| \quad (13)$$

Where the salient map S and ground truth G are normalized to the values between 0 and 1 for pixel enhancement, and w and h represent the width and height of an input frame, respectively.

1) *Quantitative Analysis*: The Quantitative and Qualitative part shows the performance comparison with existing models through F-Measure and MAE for video stream object detection. Table I. show the ability of implemented FRCNN-ResNet50 on DAVIS dataset respectively.

TABLE. I. QUANTITATIVE ANALYSIS OF PROPOSED METHOD WITH EXISTING METHODS ON DAVIS DATASET.

Methods	DAVIS Dataset		
	F-Measure	S-Measure	MAE
RFCN	0.731	0.681	0.068
SFCN	0.749	0.703	0.055
SCNN	0.774	0.739	0.074
FGRNE	0.786	0.816	0.043
Proposed FRCNN-ResNet50	0.893	0.859	0.016

2) *Comparative Analysis*: The implemented model was evaluated through a series of tests. Among the suggested models in the test set, CSAtt-ConvLSTM, Deep-GCN, and CFCN-MA achieved F-Measure rates of 0.817, 0.844, and 0.867 and MAE of 0.024, 0.031, 0.020 respectively.

TABLE. II. COMPARATIVE ANALYSIS WITH EXISTING MODELS

Methods	F-Measure	MAE
CSAtt-ConvLSTM [8]	0.817	0.024
DEEP-GCN [9]	0.844	0.031
CFCN-MA [10]	0.867	0.020
Proposed FRCNN-ResNet50	0.893	0.016

The implemented model achieved a superior detection rate, leveraging ResNet50 features. The comparison of detection F-Measure is illustrated in Table II, including models [8], [9], and [10], all trained on the DAVIS dataset.

B. Discussion

The performance comparison of the model with existing problems in video stream object detection is discussed in this section. The existing models perform various DL models, which are unable to detect the noise objects in the video stream due to the poor quality of frames, such as pose changes, occlusions, video defocusing, and motion blur. So, to resolve the above-mentioned problem, the proposed FRCNN-ResNet50 will achieve accurate positioning efficiency and reduce fault detection. The RPN is added to generate the object proposal for CNN as an input. And CNN can classify the proposal with the help of RPN input. Finally, ResNet is included to extract the feature for detecting video stream objects. Then the proposed model overcomes the disadvantages by reducing the false detection rate with the help of the ResNet50 deep network architecture. The existing models suffer from having to be trained from scratch, but the model ResNet50 is implemented in parallel and has five fully connected layers in the Faster R-CNN network to avoid false detection of video stream objects.

VII. CONCLUSION

Various CNN techniques are performed in video stream object detection that are unable to detect objects in the video stream. Due to poor quality of frames such as pose changes, occlusions, video defocusing, and motion blur problems, this research uses a FRCNN-ResNet50 to achieve accurate detection of objects in a video stream. By using the RPN to generate adaptable proposals in the DAVIS dataset, features can be used to identify the objects. Then CNN used the object features as an input for better classifying the features, which is given for feature extraction. Finally, the ResNet50 architecture extracts the essential features with the help of the Faster R-CNN model. The proposed model outperforms with an F-measure of 0.893, S-Measure and a MAE of 0.013, respectively, compared to existing models like the CSAtt_ConvLSTM, Deep-GCN, and CFCN-MA models. In the future, investigate the model to detect a specific object to improve model detection efficiency for singular objects in a video stream.

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